

江苏大学全英文授课精品课程

Area between Curves

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Review

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \tan x dx = -\ln |\cos x| + C$$

$$\int \cot x dx = \ln |\sin x| + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \tan x \sec x dx = \sec x + C$$

$$\int \cot x \csc x dx = -\csc x + C$$

$$\int x^\mu dx = \frac{1}{\mu+1} x^{\mu+1} + C, (\mu \neq -1)$$

$$\int x^{-1} dx = \int x^{-1} dx = \ln |x| + C$$

$$\int a^x dx = \frac{1}{\ln a} a^x + C; \quad \int e^x dx = e^x + C$$

$$\int 1 dx = x + C;$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C = -\arccos x + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C = -\operatorname{arccot} x + C$$

$$\int [af(x) + bg(x)] dx = a \int f(x) dx + b \int g(x) dx$$

Substitution Rule: $\int f(g(x)) \cdot g'(x) dx = \left[\int f(u) du \right]_{u=g(x)}$

Substitution Rule in Inverse: $\left[\int f(u) du \right]_{u=g(x)} = \int f(g(x)) \cdot g'(x) dx$

By Parts: $\int f(x) \cdot g'(x) dx = f(x) \cdot g(x) - \int f'(x) \cdot g(x) dx$

TABLE OF TRIGONOMETRIC SUBSTITUTIONS

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$





$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x$$

FTC 2

If f is a continuous function on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Where F is any antiderivative of f , that is, a function such that

$$F' = f$$

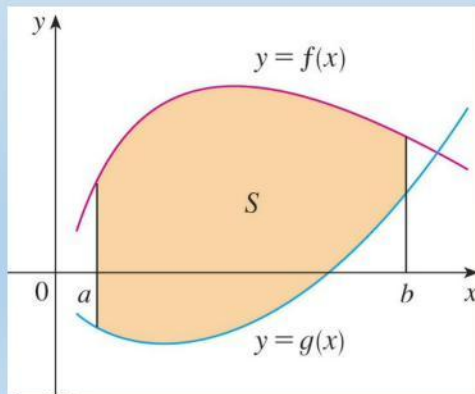




1 Properties of the Integral

We Consider the region S that lies between two curves $y = f(x)$ and $y = g(x)$ and between the vertical lines $x = a$ and $x = b$.

Here, f and g are continuous functions and $f(x) \geq g(x)$ for all x in $[a, b]$.



Divide $[a, b]$: $a = x_0 < x_1 < \cdots < x_n = b$, $\Delta x = x_i - x_{i-1} = (b - a)/n$.

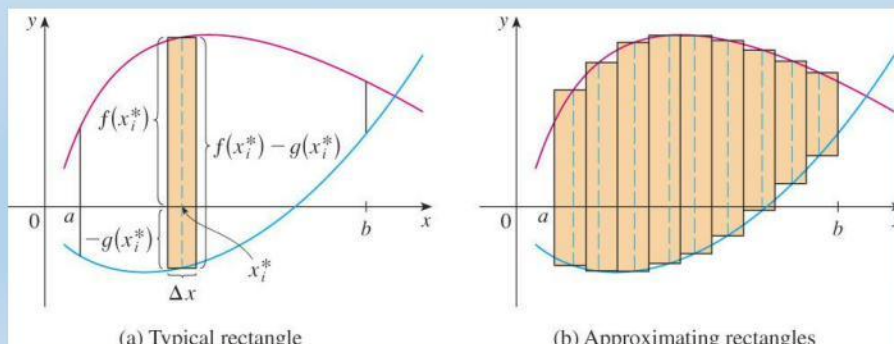
$$\Delta S_i \approx (f(x_i^*) - g(x_i^*))\Delta x$$

Take a sum:

$$S = \sum_{i=1}^n \Delta S_i \approx \sum_{i=1}^n (f(x_i^*) - g(x_i^*))\Delta x$$

Take a limit:

$$S = \lim_{n \rightarrow \infty} \sum_{i=1}^n (f(x_i^*) - g(x_i^*))\Delta x = \int_a^b (f(x) - g(x))dx$$





The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, is:

$$A = \int_a^b [f(x) - g(x)] dx$$

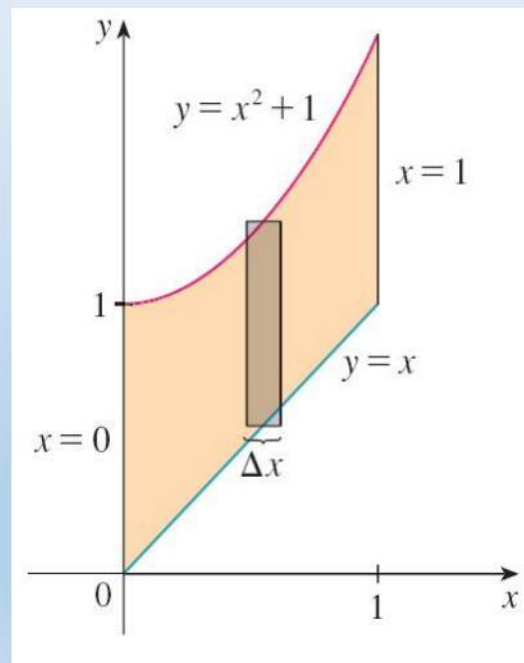
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Example 1 Find the area of the region bounded by $y = x^2 + 1$, $y = x$, $x = 0$, $x = 1$.

Solution:

$$\begin{aligned} A &= \int_0^1 \left[(x^2 + 1) - x \right] dx \\ &= \int_0^1 (x^2 - x + 1) dx \\ &= \left[\frac{x^3}{3} - \frac{x^2}{2} + x \right]_0^1 = \frac{1}{3} - \frac{1}{2} + 1 = \frac{5}{6} \end{aligned}$$



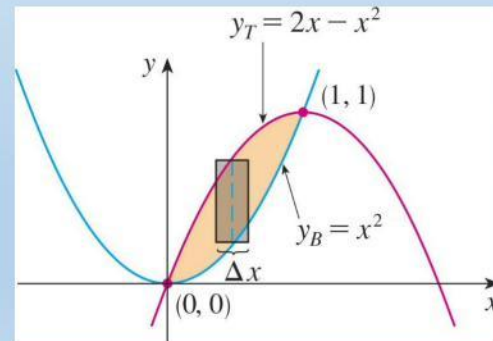
Example 2 Find the area of the region bounded by $y = x^2$, $y = 2x - x^2$.

Solution: First, we find the points of intersection of the parabolas by solving their equations simultaneously.

$$x^2 = 2x - x^2 \Rightarrow x = 0 \text{ or } x = 1$$

The points of intersection are $(0, 0)$ and $(1, 1)$.

$$\begin{aligned} A &= \int_0^1 (2x - 2x^2) dx = 2 \int_0^1 (x - x^2) dx \\ &= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3} \end{aligned}$$





Exercises

Evaluate the area of the regions bounded by the given curves :

(a) $y = x^2 - 1$, $y = e^x$, $x = -1$, $x = 1$

(b) $y = x^2 - 2x$, $y = x + 4$

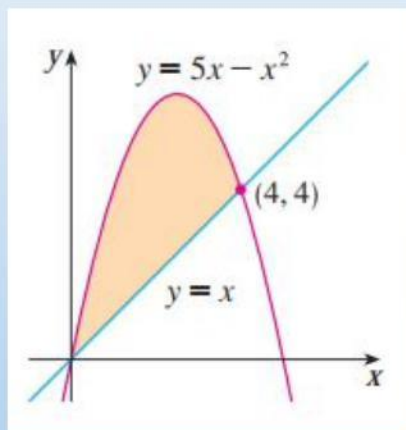
(c) $y = \sqrt{x}$, $y = \frac{1}{2}x$,

(d) $y = \cos x$, $y = \sin 2x$, $x = 0$, $x = \frac{\pi}{2}$

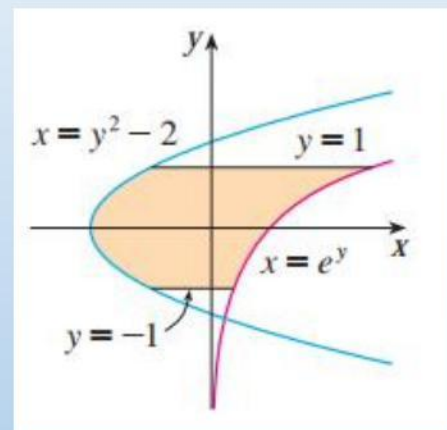


Exercises 2 Find the area of the shaded region:

(a)



(b)





Exercises 3

Sketch the region enclosed by the given curves and find its area:

(a) $y = 12 - x^2$, $y = x^2 - 6$

(b) $y = \cos x$, $y = 1 - \cos x$, $0 \leq x \leq \pi$

(c) $y = \sqrt{x}$, $y = x^2$

(d) $y = |x|$, $y = x^2 - 2$, $x = 0$, $x = 2$

